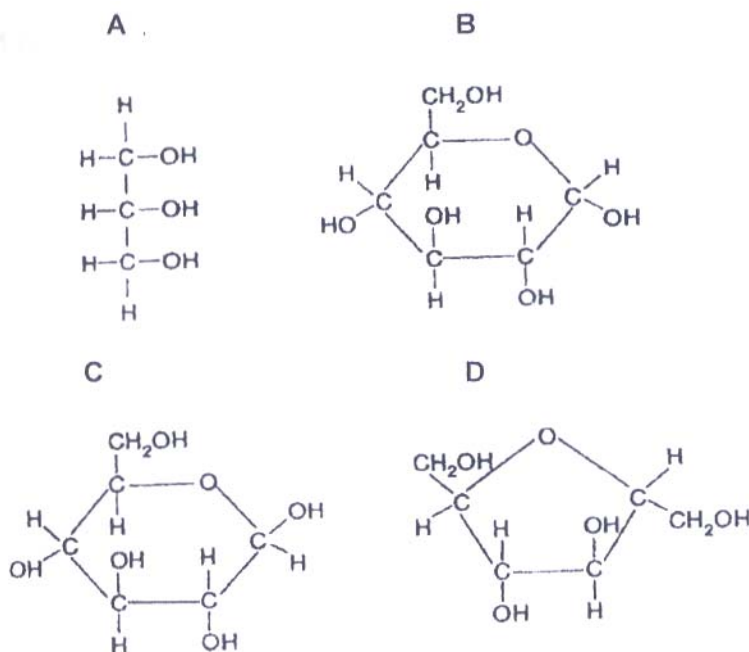


1 The following statements refer to the presence of compounds such as phospholipids, RNA and DNA in all or some of the organelles. Choose the most appropriate option.

- A Ribosomes are the only organelles that have all the three compounds
- B Both smooth endoplasmic reticulum and mitochondria have only phospholipids
- C Both ribosomes and mitochondria have phospholipids and DNA
- D Mitochondria are the only organelles that have all the three compounds

2 Which molecule is found in starch?



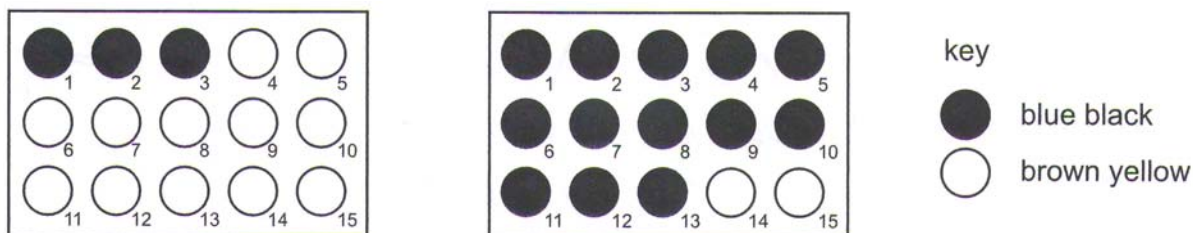
3 The primary structure of a molecule of enkephalin, a mammalian neurotransmitter, is shown.

Tyrosine – glycine – glycine – phenylalanine – leucine

Which combination correctly describes the synthesis and structure of one molecule of enkaphalin?

	Number of molecules of water eliminated	Type of bond	Number of different R groups in molecule
A	4	Glycosidic	5
B	4	Peptide	4
C	5	Glycosidic	5
D	5	Peptide	4

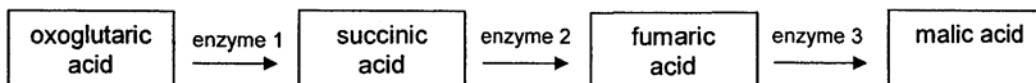
- 4 An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and placed onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.



Which shows the correct temperatures and times for the complete digestion of starch?

	Temperature/ °C	Time/s
A	30	3.15
	10	0.45
B	30	45
	10	195
C	10	45
	30	195
D	10	3.15
	30	0.45

- 5 The four acids shown below form part of an enzyme-catalyzed sequence.

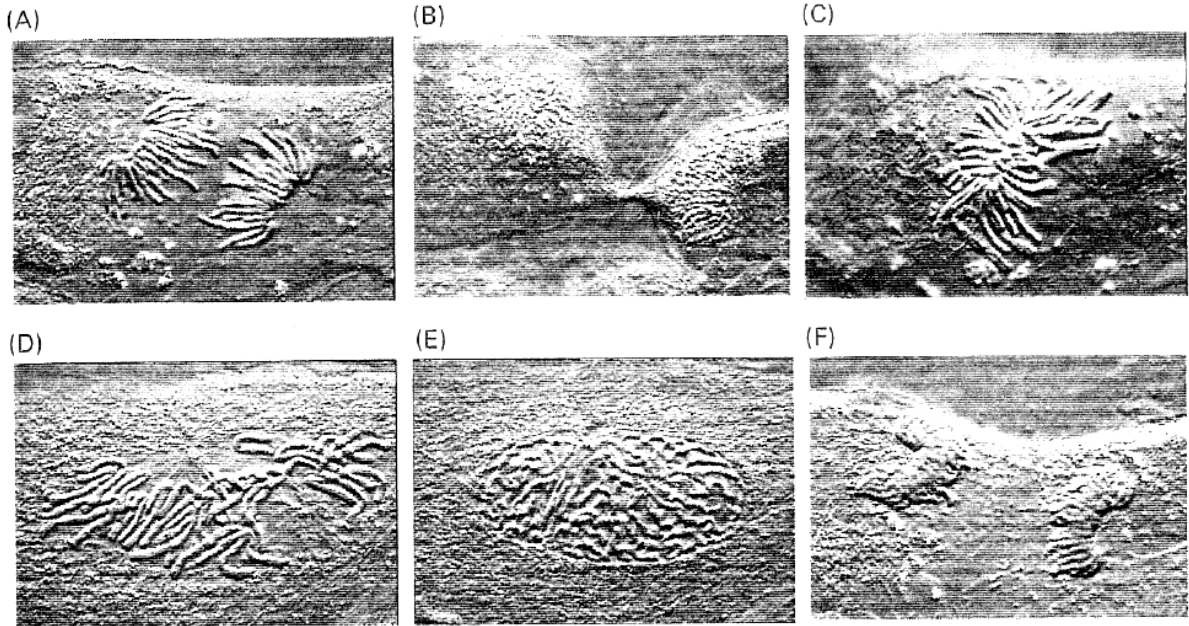


The addition of malonic acid results in no change in the concentration of oxoglutaric acid, an accumulation of succinic acid, and a near absence of both fumaric acid and malic acid. Further addition of fumaric acid results in the formation of malic acid.

What does this information indicate about malonic acid?

- A** It is an inhibitor of enzyme 2.
- B** It catalyzes the formation of succinic acid.
- C** It is an inhibitor of enzyme 1.
- D** It reacts with fumaric acid.

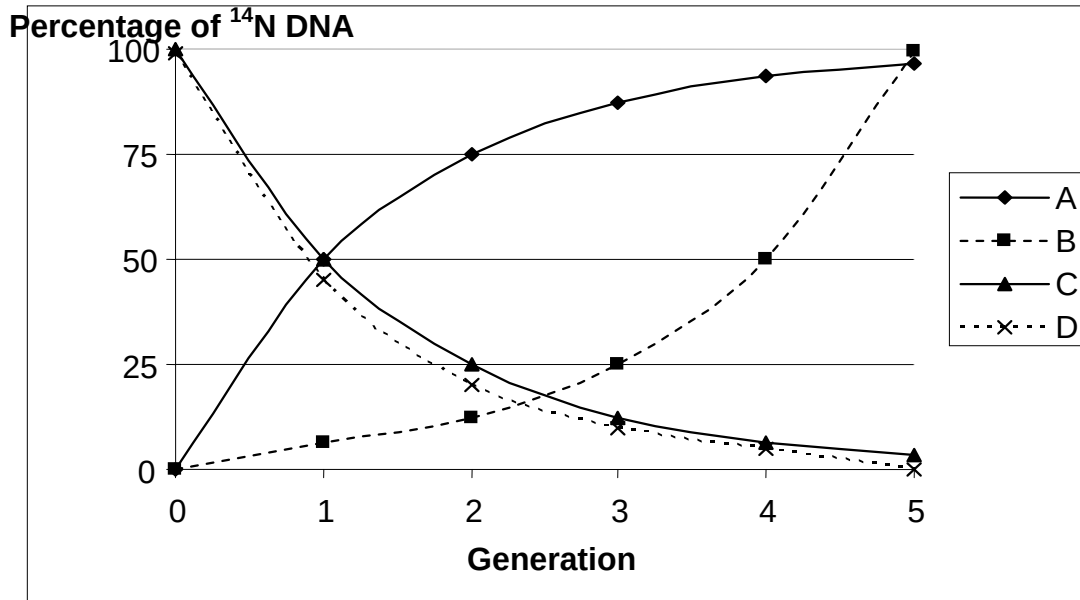
- 6 A living cell from the lung epithelium of a newt is shown at different stages of mitosis in the micrographs (**P** to **U**) below. Identify the correct sequence of these micrographs.



- A ECDAFB
 B DECAFB
 C EDCAFB
 D EDCBAF
- 7 In a hypothetical species of frogs, there are 6 tetrads at metaphase of meiosis I. How many chromosomes would be present in each cell at the end of meiosis II?
- A 3
 B 6
 C 12
 D 24

- 8** The protein, RB, is a tumour suppressor encoded by a gene on chromosome 13 of the human genome. Which individual is most likely to develop the associated cancer, retinoblastoma?
- A** Has a frameshift mutation of the RB allele on each chromosome 13.
 - B** Has an RB allele on each chromosome 13.
 - C** Has an RB allele on one chromosome 13.
 - D** Has an RB allele on one chromosome 13 and another, by translocation, on one chromosome 6.
- 9** Which of the following statements about the structure of DNA is incorrect?
- A** One complete turn requires 3.4nm and 10 base pairs.
 - B** The backbones of each strand run in opposite directions relative to each other.
 - C** Each pair of nucleotides is held together by three hydrogen bonds.
 - D** The width of the molecule is a constant 2nm.
- 10** In studying a virus, you find the following proportions of nitrogenous bases present: adenine 23%, guanine 31%, cytosine 23%, unexpected base 17%. What statement(s) can you make about this virus?
- A** It probably uses RNA as its genetic material.
 - B** The genetic material of this virus is probably single stranded.
 - C** Base pairing rules in this virus probably include adenine:cytosine.
 - D** A and B

- 11 Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all DNA was labelled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for 5 generations. The percentage of ^{14}N DNA strands in each generation was estimated. Which curve provides evidence that DNA replication is semi-conservative?



- 12 A mutation in one of the replication enzymes causes DNA polymerase to be unable to add nucleotides at the origin of replication, hence no daughter strands of DNA can be synthesized. Which of the following is the defective enzyme?

- A Helicase
- B Primase
- C DNA Ligase
- D Topoisomerase

- 13 In most organisms, all possible triplets of bases in DNA code for an amino acid, except for the coding strand 'stop' triplets ATT, ATC and ACT. The coding strand is complementary to mRNA.

In the prokaryote, *Methanosarcina barkeri*, the coding strand triplet ATC codes for an unusual amino acid pyrrolysine.

The **non-coding** strand of a section of DNA in *M. barkeri* has the following sequence of bases.

TTTTTATTGTATTACTAGTGTTAATGA

How many amino acids can be made into a peptide by *M. barkeri* from the **coding** strand of this region of DNA?

- A 5
- B 6
- C 7
- D 9

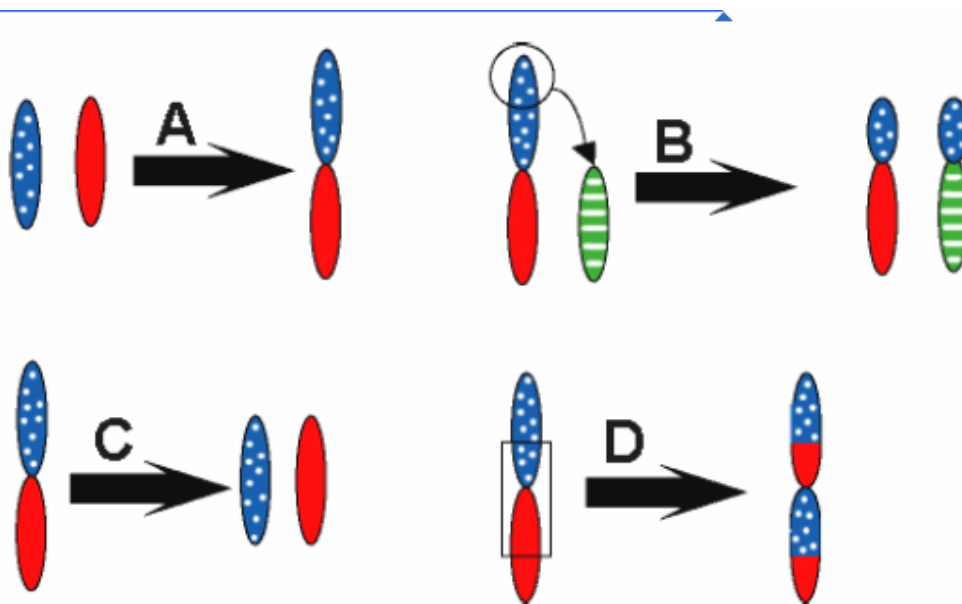
14 The list shows the stages in the cellular synthesis of proteins.

1. movement of mRNA from nucleus to cytoplasm
2. linking of adjacent amino acid molecules
3. transcription of mRNA from a DNA template
4. formation of the polypeptide chain
5. attachment of the mRNA strand to a ribosome

Which sequence is correct?

- A** $1 \rightarrow 3 \rightarrow 2 \rightarrow 5 \rightarrow 4$
- B** $1 \rightarrow 5 \rightarrow 3 \rightarrow 4 \rightarrow 2$
- C** $3 \rightarrow 1 \rightarrow 5 \rightarrow 2 \rightarrow 4$
- D** $3 \rightarrow 4 \rightarrow 1 \rightarrow 2 \rightarrow 5$

15 Which of the following illustrates chromosomal inversion?



16 A virus has a base ratio of $(A + G) / (U + C) = 1$. What type of virus is this?

- A** a single-stranded DNA virus
- B** a double-stranded DNA virus
- C** a single-stranded RNA virus
- D** a double-stranded RNA virus

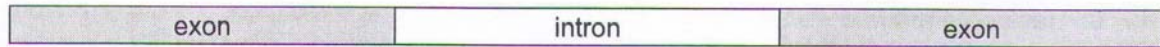
- 17** A group of scientists uses modified viruses as vectors for the introduction of foreign genes into bacteria.
In one experiment, a bacteriophage was put together with the protein coat of T4 phage and the DNA of a lambda phage. If this composite phage was allowed to infect bacteria, the phages produced in the bacteria will have
- A** The protein of lambda phage and the DNA of T4.
 - B** The protein of T4 and the DNA of lambda phage.
 - C** The protein and DNA of lambda phage.
 - D** The protein and DNA of T4.
- 18** A mutant strain of *E. coli* has been isolated in which the *lac* operon is not expressed in the presence of lactose. To understand the nature of this defect, this parent strain was mated so that it now contains an F' factor containing a normal *lac* operon. The parent and mated strain with regard to their beta-galactosidase activities in the presence and absence of lactose was compared. The following results were obtained:

Strain	Addition of lactose	Amount of beta-galactosidase (percentage of parent strain)
Parent	No	0
Parent	Yes	0
Mated	No	0
Mated	Yes	100

With respect to the results shown in table, which part of the operon most likely has the defect/mutation?

- A** *lac A*
 - B** *lac I*
 - C** Promoter
 - D** Operator
- 19** Which of the following would you expect to find in an inducible system?
- A** A repressor protein, which is bound to DNA in the absence of an inducer.
 - B** A repressor protein, which is bound to DNA in the presence of a corepressor.
 - C** An activator protein, which is bound to DNA in the absence of an inducer.
 - D** An activator protein, which is bound to DNA only in the absence of an inhibitor.

20 The diagram shows a gene in a human cell.



In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

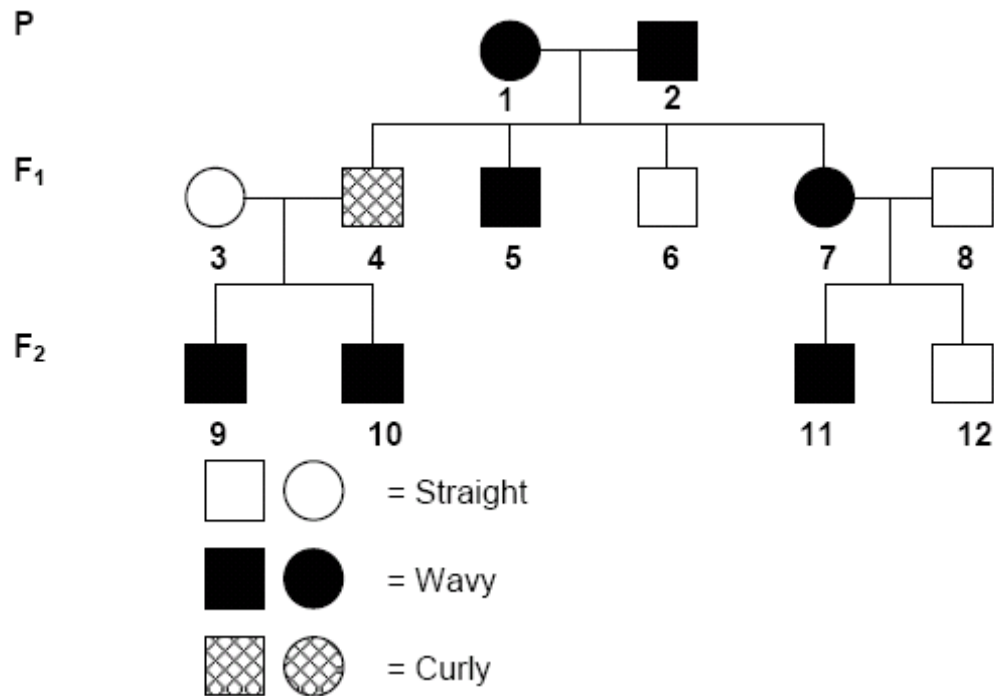
Which stage of gene expression would this mutation directly affect?

- A** Activation of the gene by a steroid hormone
- B** Attachment of mRNA to a ribosome
- C** Binding of RNA polymerase to DNA
- D** Splicing of pre-mRNA

21 What is an example of translational control of gene expression?

- A** Activation of proteins by folding or enzymatic cleavage
- B** Addition of chemical groups, such as phosphate groups, to free amino acids in the cytoplasm
- C** Binding of protein factors to specific sequences in mRNA preventing ribosomes attaching
- D** Formation of disulphide bridges in the protein being formed.

Questions 22 and 23 refer to the following pedigree chart.



22 Which of the following statements is true?

- A** Wavy hair is dominant.
- B** Individual 1 is homozygous.
- C** Individual 4 is homozygous.
- D** Curly hair is recessive to straight hair.

23 Which of the following statements has a 50% chance of being true?

- I** One of the parents of individual 2 had the same phenotype as individual 2.
 - II** If individual 10 married someone with wavy hair, the first child would have wavy hair.
 - III** If individual 6 married a woman with straight hair, all of the offspring would have straight hair.
- A** II only
 - B** I and III only
 - C** II and III only
 - D** All of the above

- 24** It has been found that an aqueous suspension of isolated chloroplasts will evolve oxygen if illuminated in the presence of a certain type of compound.

Which type of compound and which colours of light are required for maximum oxygen evolution?

	<i>type of compound</i>	<i>colours of light at which maximum evolution occurs</i>
A	electron acceptor	blue and green
B	electron acceptor	blue and red
C	electron donor	blue and green
D	electron donor	blue and red

- 25** Dinitrophenol is a metabolic poison that can lodge within the thylakoid membranes of chloroplasts. It then provides an alternative route for H^+ ions to diffuse across the thylakoid membranes. In what way will the Calvin cycle (dark reaction) be affected in chloroplasts poisoned with dinitrophenol?

- A** No effect since Calvin cycle is an enzyme-controlled process.
- B** The rate of Calvin cycle will increase because pH in the stroma will drop towards the optimum.
- C** The rate of Calvin cycle will decrease with the accumulation of glycerate-3-phosphate (GP).
- D** The rate of Calvin cycle will decrease with the accumulation of glyceraldehyde-3-phosphate (G3P) i.e. triose phosphate (TP).

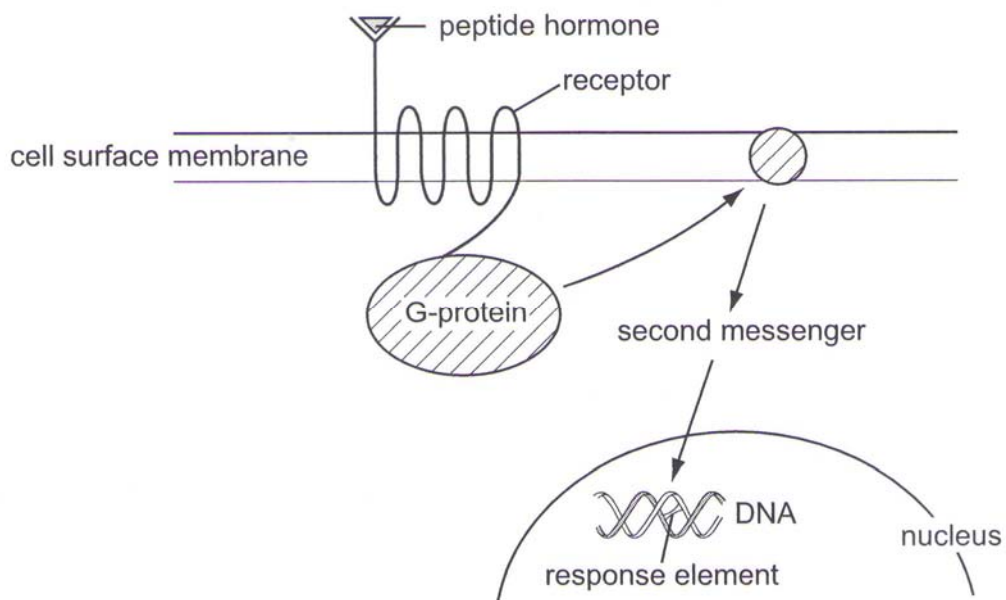
- 26** When cyanide is bound to cytochrome oxidase, the cell can no longer aerobically produce ATP for energy. Which statement below best explains this?

- A** It prevents photolysis from occurring to produce oxygen.
- B** It prevents NADPH from being oxidised to $NADP^+$, hence preventing electron flow down the electron transport chain. No chemiosmosis occurs.
- C** It prevents pyruvate from being synthesised in glycolysis, hence stopping the Krebs cycle.
- D** It prevents oxygen from accepting electrons and protons to form water, hence preventing electron flow down the electron transport chain. No chemiosmosis occurs.

- 27** A young boy never had much energy. He goes to a doctor for help and is sent to the hospital for some tests. There they discover his mitochondria can use only fatty acids and amino acids for respiration, and his cells produce more lactate than normal. Of the following, which is the best explanation of his condition?
- A** His cells cannot move NADH from glycolysis into the mitochondria.
 - B** His mitochondria lack the transport protein that moves pyruvate across the outer mitochondrial membrane.
 - C** His cells contain something that inhibits oxygen use in his mitochondria.
 - D** His cells have a defective electron transport chain, so glucose goes to lactate instead of to acetyl CoA.
- 28** Which of the following statements **BEST** describes the outcome of the negative feedback loop in a homeostatic mechanism?
- A** Glucagon is released by the pancreas in response to a drop in blood glucose concentration below the normal level.
 - B** The liver cells break down glycogen to glucose in response to glucagon.
 - C** Blood glucose level increases to the normal level in response to the release of glucagon.
 - D** Release of glucagon is stopped in response to blood glucose concentration returning to the normal level.
- 29** Which of the following statements correctly describes the effect of blood insulin?

	insulin level in blood	Effect
A	high	a decrease in glucose metabolism
B	high	an increase in conversion of glycogen to glucose
C	low	a decrease in glucose permeability of cells
D	low	an increase in uptake of glucose by muscle cells

- 30 Which of the following offers the best description of neural transmission across a mammalian synaptic gap?
- A Neural impulses involve the flow of K^+ and Na^+ across the gap.
 - B Neural impulses travel across the gap as electrical currents.
 - C Neural impulses cause the release of chemicals that diffuse across the gap.
 - D Neural impulses travel across the gap in both directions.
- 31 The diagram represents a G-protein coupled receptor on the cell surface membrane. A peptide hormone ligand is bound to the receptor and initiates the production of a second messenger.



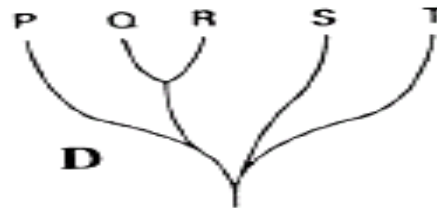
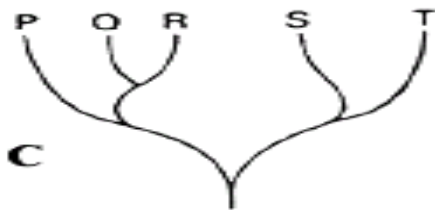
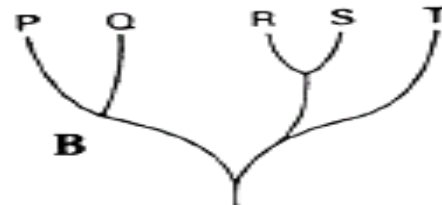
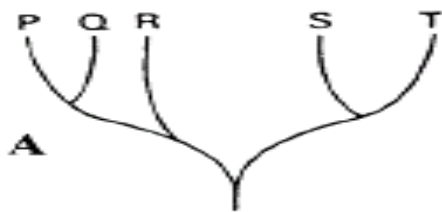
What is the second messenger?

- A A peptide hormone
- B ATP
- C Cyclic AMP
- D Kinase

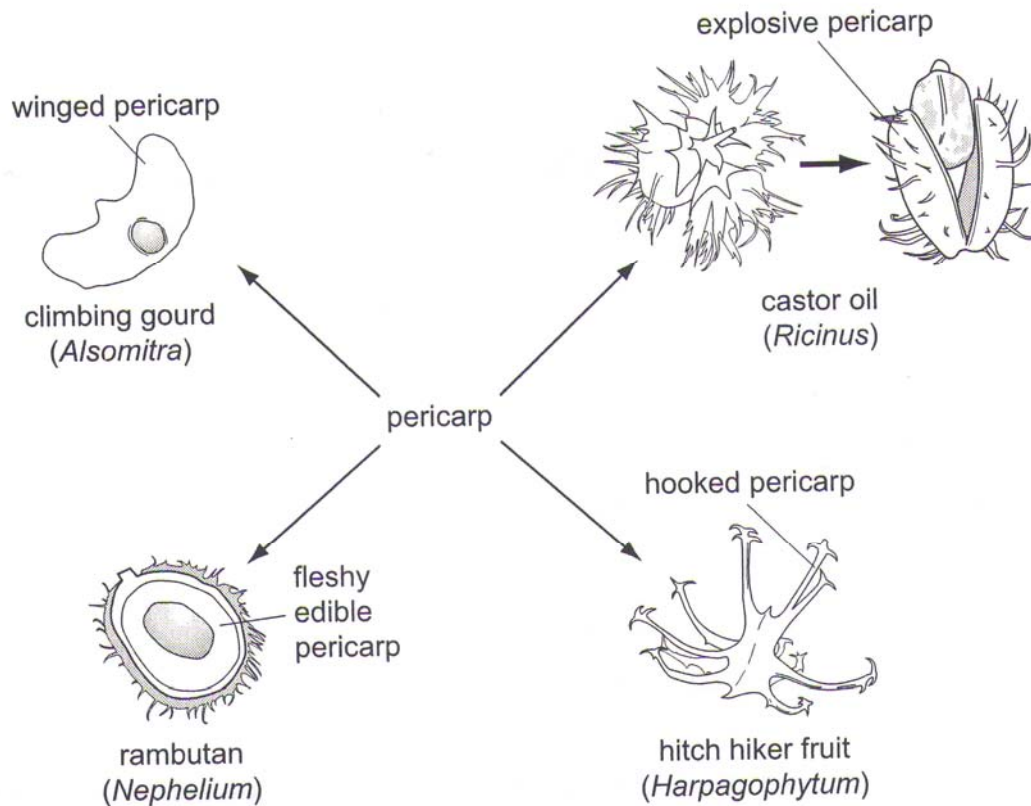
- 32** Five species, **P, Q, R, S and T**, possess the protein cytochrome *c* oxidase. The primary structure of the protein was determined for each species and the number of amino acid differences between species is given below.

	P	Q	R	S	T
P	0	7	8	20	22
Q	7	0	3	19	17
R	8	3	0	18	21
S	20	19	18	0	10
T	22	17	21	10	0

From the evidence, which of the following represents the probable evolutionary relationship between the species?



- 33 The diagram illustrates variation in the pericarp (fruit wall) for a variety of methods of seed dispersal.



What do these examples illustrate?

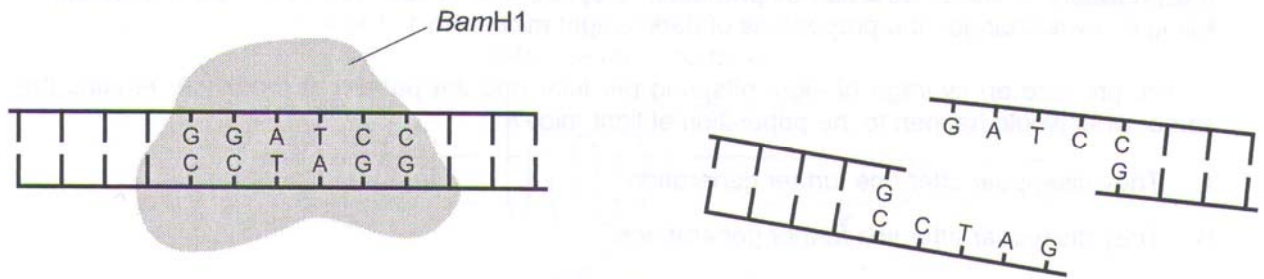
- A The adaptive radiation of analogous structures showing convergent evolution.
- B The adaptive radiation of analogous structures showing divergent evolution.
- C The adaptive radiation of homologous structures showing convergent evolution.
- D The adaptive radiation of homologous structures showing divergent evolution.

- 34** Molecular homology of haemoglobin is used to determine evolutionary relatedness in a range of mammals. Either nucleic acids or proteins can be sequenced. The sequence of nucleic acids and proteins are compared.

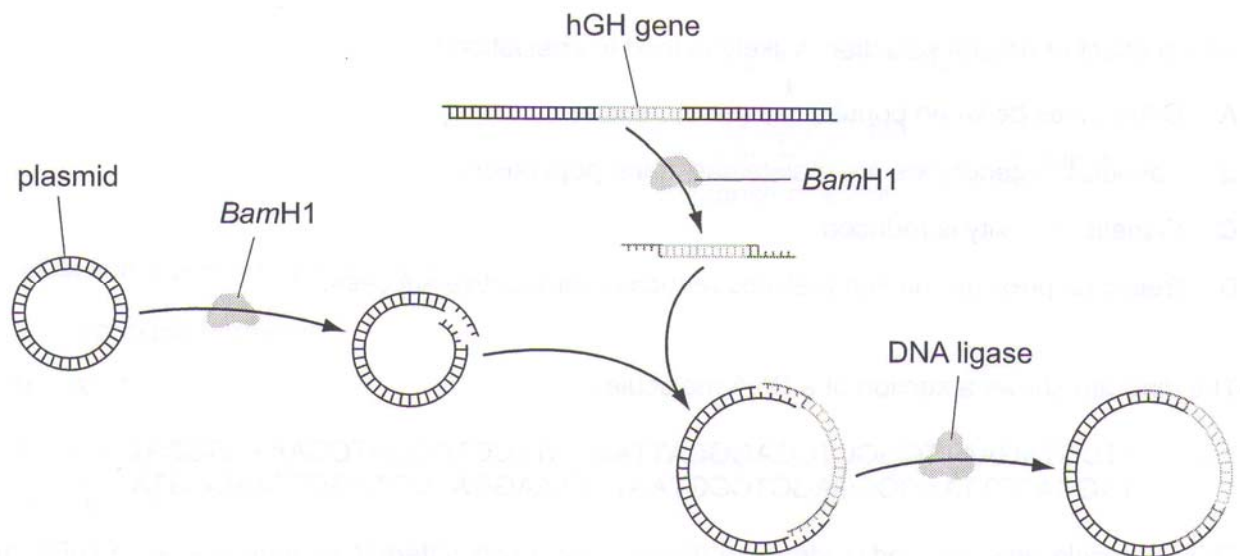
Which sequence will vary **least** in related mammals?

- A** Amino acid
 - B** DNA
 - C** Edited mRNA transcript
 - D** Primary mRNA transcript
- 35** Assume that you are trying to insert a gene into a plasmid. Someone gives you a preparation of genomic DNA that has been cut with restriction enzyme X. The gene you wish to insert has sites on both ends for cutting by restriction enzyme Y. You have a plasmid with a single site for Y, but not for X. Your strategy should be
- A** insert the fragments cut with X directly into the plasmid without cutting the plasmid
 - B** cut the plasmid with restriction enzyme X and insert the fragments cut with Y into the plasmid
 - C** cut the DNA again with restriction enzyme Y and insert these fragments into the plasmid cut with the same enzyme
 - D** cut the plasmid twice with restriction enzyme Y and ligate the two fragments onto the ends of the DNA fragments cut with restriction enzyme X

- 36 BamH1 is a restriction enzyme that cuts DNA as shown in the diagram.



The diagram below shows part of the procedure for producing *E. coli* that will synthesize human growth hormone, hGH.



At the end of this process, many plasmids do not contain the hGH gene.

What could explain this?

- A Different alleles of the hGH gene have different sticky ends.
- B Not all of the plasmids cut by BamH1 have sticky ends.
- C Some of the plasmids are cut at more than one position.
- D The sticky ends of some of the plasmids rejoin with each other.

- 37** A student performed the following steps in a Southern blot experiment to determine the number of copies of a particular gene that has been inserted in a genetically modified organism.
- Transfer of DNA to nitrocellulose membrane.
 - Restriction digestion of genomic DNA.
 - Cleaved DNA separated using gel electrophoresis.
 - Create radioactive probe.
 - Incubate probe and membrane.

Which is the correct sequence to the above steps?

- ii → iii → i → iv → v
 - ii → iii → i → v → iv
 - iv → v → i → ii → iii
 - v → iv → iii → ii → i
- 38** How does a genomic library differ from a cDNA library?
- A genomic library contains only noncoding sequences, whereas a cDNA library contains only coding sequences.
 - A genomic library varies, dependent on the cell type used to make it, whereas the content of a cDNA library does not.
 - A genomic library can be made using a restriction enzyme and DNA ligase only, whereas a cDNA library requires both of these as well as reverse transcriptase and DNA polymerase.
 - The genomic library contains only the genes that can be expressed in the cell
- 39** Liposuction waste has been reported to contain 50-100 million stem cells per 250 g. When these stem cells were harvested, they could not differentiate into any other cell except for fat, bone and cartilage.

Which of the following statements best explains this?

- It is a multipotent adult stem cell which can only specialise into limited cell types.
- It is a multipotent embryonic stem cell which can specialise into almost all the cell types.
- It is a pluripotent embryonic stem cell which can specialise into almost all the cell types.
- It is a pluripotent adult stem cell which can specialise into almost all the cell types.

40 Which process could produce a large number of genetically identical plants?

- A** Anther culture
- B** Embryo culture
- C** Pollen culture
- D** Protoplast culture